Supplementary Material: Stress-Dependent Cardinal Boundaries of Native Assignment States in Aerial Tiny-Object Detection

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1. Evidence hierarchy and interpretation rules

This Supplementary Material expands the frozen evidence without adding training runs or post hoc outcome selection. Authority descends from `evidence_freeze/final_evidence_v4` to the current ledger, historical factual records, and manuscript prose. The v4 bundle preserves `final_evidence_v3` byte for byte. Its manifest SHA-256 is

`00513d016526909f2c431df12b056f93aa1ea83756497f385caca6164c01cb24`.

The v4 ledger contains 32 admitted records and five excluded historical records; no scientific gate is pending. Renewed author approval and release synchronisation remain operational gates.

Principal endpoints are inverse-inclusion weighted. Unless stated otherwise, 95% intervals use 5,000 image-cluster bootstrap replicates within frozen strata and quantify sampling uncertainty conditional on seed 0. Discovery and sensitivity analyses were not a confirmatory family and received no family-wise multiplicity correction. Fixed and equivalent-area-side contracts are not pooled. First-divergence labels localise the earliest instrumented change. The lockbox produced no normative verdict; post-access corrections remain exploratory.

Table S1: Evidence-status vocabulary used throughout the paper.

Status	Meaning
Qualified instrument	Analytical oracles, mutation tests, invariants, native parity, and independent-reference checks passed.
Discovery estimate	Frozen outcome-blind discovery-sample estimate with prespecified weighting and cluster bootstrap.
Sensitivity/replication	Read-only analysis under another contract, dataset, checkpoint, or state definition.
Negative boundary	Prespecified analysis did not establish the claimed utility or treatment effect; not an equivalence claim.
No normative verdict	The single lockbox analyzer invocation ended before any preregistered estimate or decision was written.
Post-access exploratory	Corrective estimate computed after outcome access; formal status is permanently null.

2. Detector, dataset, sampling, and replay contracts

The principal contract was YOLO26s trained on AI-TOD-v2 with native O2O and O2M branches; VisDrone and compatible YOLOv10-S analyses are sensitivity/replication records. Detector outputs were computed once and held fixed during GT-only replay, which moved only the focal GT centre. Source paths and checkpoint hashes are tabulated below; raw candidate tensors are not in the public release.

The AI-TOD-v2 discovery bundle contained 300 outcome-blind stratified validation images. The frozen population-facing scale domain was $s_g = \sqrt{w_g h_g} \in [8, 32)$ px, partitioned at 16 px and shared with the VisDrone sampling design. Objects below 8 px were outside these outcome strata and were not retrospectively added.

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Sampling weights were inverse image-inclusion probabilities. The principal equivalent-area-side displacement was $\delta_g = 0.0625\sqrt{w_g h_g}$, with $\kappa \in \{0.03125, 0.0625, 0.125\}$ used for stress-magnitude sensitivity. This scales by the side length of an equal-area square; for rectangular GTs it does not impose equal fractions of width and height. Invalid boundary shifts were skipped rather than clipped.

The executable eligibility, alignment, Top-k, conflict, tie, and final-state rules are given in the principal contract table below; they were read directly from the frozen assigner and applied in native float32.

Table S2: Principal YOLO26s executable native-state contract. Operations are applied in the listed order; the detector-specific YOLOv10-S Top-k difference is reported separately below.

Stage	O2O	O2M
Eligibility	Replace each GT dimension < 8 px by 16 px for the eligibility rectangle; retain anchor centres with all signed edge distances $> 10^{-9}$.	Identical geometric rule on the O2M candidate grid.
Alignment	$q = p_{c_g}^{0.5} \max\{\mathrm{CIoU}, 0\}^6$ in native float32.	Same formula using the native O2M branch scores and decoded boxes.
Preselection	Native `topk` with $k = 7$.	Native `topk` with $k = 10$.
Multi-GT claim	For every multiply claimed anchor, choose the GT with maximal clipped CIoU using native `argmax`.	Identical conflict operation.
Final state	Apply native Top-1-by-q among surviving claims; read the unique post-conflict assigned candidate from `fg_mask` and `target_gt_idx`. Positive normalised `target_scores` are required separately for nonzero box/DFL weight.	Retain the complete post-conflict positive set; assigned-set Top-1 is its maximal-q member.
Undefined/empty	Missing unique post-conflict assigned O2O identity or missing positive alternative is undefined; no extreme-value imputation.	Empty/empty, empty/nonempty, and nonempty/empty are explicit structural states; Top-1 is undefined for an empty set.
Tie order	Exact ties follow native PyTorch `topk`/`argmax`; no secondary symbolic tie rule is introduced.	Identical native ordering contract.

The registered checkpoint hash prefixes are `4b57787f7351` for YOLO26s and `ecf7ca8df8ed` for YOLOv10-S. Their complete SHA-256 values are retained in `MODEL_ARTIFACTS_SHA256.csv` and the detector-specific manifests. The YOLOv10-S extraction manifest (SHA-256 prefix `c1e14e823fab`) records the detector contract and checkpoint but not a contemporaneous source commit.

Table S3: Detector-contract comparison. Both contracts use native float32 ordering, the same task-alignment form, and maximal-clipped-CIoU resolution of multiply claimed candidates, but their native O2O preselection cardinalities differ. Assigned-set Top-1 is an O2M rank-state probe, not a uniquely supervised positive.

Detector	Released source path	O2O selection	O2M selection	Audited final state
YOLO26s	`ultralytics/utils/loss.py`: `Detect`	native Top-7, then Top-1 survivor	native Top-10	unique post-conflict assigned O2O identity from `fg_mask` and `target_gt_idx`; checkpoint SHA-256 `4b57787f7351...`
YOLOv10-S	`ultralytics/utils/loss.py`: `v10Detect`	native Top-1	native Top-10	O2O: post-conflict assigned identity under native Top-1 contract; O2M: complete post-conflict positive set and maximal-q Top-1 rank probe; checkpoint SHA-256 `ecf7ca8d...`

The VisDrone row is case B: a VisDrone-trained YOLO26s checkpoint underwent the same focal-GT-only replay; the provenance table distinguishes training data, roster, and common-valid support.

Table S4: Cross-dataset and detector provenance for the audited replay results. Image counts are selected/contributing; GT support is the common-valid replay support. All rows use outcome-blind stratified image selection and inverse image-inclusion weights. The VisDrone row is a VisDrone-trained checkpoint (case B), not an AI-TOD-trained checkpoint replayed on VisDrone.

Dataset / analysis	Detector	Training dataset	Checkpoint role	Seed	Selected / contributing images	GT support	Replay contract	Checkpoint SHA-256
AI-TOD-v2 primary	YOLO26s	AI-TOD-v2	Seed-0 discovery / primary checkpoint	0	300 / 288	6,378 fixed / 6,377 equivalent-area common-valid GTs	Focal-GT-only, frozen outputs; fixed 1 px and equivalent-area-side $\kappa = 0.0625$	`4b57787f7351c77dfe6c85e64b30206245207722b7ed86cae0c741b1f06828fa`
VisDrone cross-dataset sensitivity	YOLO26s	VisDrone2019-DET	VisDrone native b4/e150 seed-0 checkpoint	0	300 / 300	13,911 common-valid GTs	Focal-GT-only, frozen outputs; equivalent-area-side $\kappa \in \{0.03125, 0.0625, 0.125\}$	`e7cffd98c6055bb0818873f94e6566feb697660ec5666f21f13c6304b412d0e8`
YOLOv10-S compatible-contract replication	YOLOv10-S	AI-TOD-v2	Frozen native best checkpoint; cross-detector replication	0	300 / 288	6,348 common-valid GTs	Focal-GT-only, frozen outputs; equivalent-area-side $\kappa = 0.0625$; native O2O Top-1 / O2M Top-10	`ecf7ca8df8eded54b9500109123f94ef8eaddd736bd8351a968761f0f62a0d38`

2.1. Training and checkpoint contract

The four protocol-eligible YOLO26s artifacts retained under the filename `best.pt` share the same training arguments apart from seed and output path. Table S5 distinguishes directly recorded values from quantities that were not preserved. The model registry and dataset-path template have SHA-256 prefixes `e8915dbf96e8` and `1f5a0e11a28c`; their complete hashes remain in the release manifest.

Table S5: Training and checkpoint contract recovered from the retained YOLO26s checkpoint metadata. Values marked unavailable were not reconstructed from the current source tree.

Field	Frozen record
Model and initialisation	`weights/yolo26s.pt`; `pretrained=true`. The initial-weight SHA-256 was not retained.
Data and split	`configs/aitod_v2.yaml`; `train/images` for training and `val/images` for formal post-training evaluation. Reported mAP values use the validation split, not the 300-image audit sample.
Image and batch contract	`imgsz=800`, `batch=8`, nominal batch `nbs=64`.
Run length and saving	`epochs=300`, `patience=50`, `save_period=20`. The retained `best.pt` files record zero-based epoch 280.
Optimisation record	`optimizer=auto`, `lr0=0.01`, `lrf=0.01`, `cos_lr=false`, `weight_decay=0.0005`. The resolved optimizer name and state were not retained and are not inferred as executed facts.
Warm-up	`warmup_epochs=3.0`, `warmup_momentum=0.8`, `warmup_bias_lr=0.0`.
Photometric augmentation	`hsv_h=0.015`, `hsv_s=0.7`, `hsv_v=0.4`.
Geometric augmentation	`translate=0.1`, `scale=0.5`, `degrees=0`, `shear=0`, `perspective=0`, `fliplr=0.5`, `flipud=0`.
Composition augmentation	`mosaic=1.0`, `close_mosaic=10`, `mixup=0`, `cutmix=0`, `copy_paste=0`.
Runtime arguments	`amp=false`, `deterministic=true`, `device=0`, `workers=8`, `cache=ram`; protocol-eligible seeds 0–3.
Checkpoint selection	Training recorded `val=false`; embedded `best_fitness` and `fitness` are null. A post-training scan on the full validation benchmark selected epoch 280 for seeds 0–2 and promoted it to the retained filename `best.pt`; the corresponding seed-3 promotion log was not recovered. Replay outcomes did not select weights.

Complete seed-0–3 hashes and the checkpoint-provenance audit are retained in the release manifest and `results/measurement_validation_20260905/checkpoint_provenance_v1/summary.json`; seed 4 was excluded before analysis because it did not use the common initialisation.

The coverage-based-miss outcome counts the absence of a frozen same-class prediction overlapping the focal GT at IoU ≥ 0.5; it is not the standard evaluator false negative.

3. Instrument qualification and numerical parity

All six analytical oracles, seven property invariants, and six prespecified mutations passed; native, traced, and independent-reference outputs matched across 300 base and 5,251 directional states.

Qualification used 30 frozen images and 4,408 trajectories; the focal-row engine later matched 3,072 full-grid scenarios byte-for-byte. The 300-image/5,251-state parity run found positive target scores for all 8,079 base and 5,159 shifted assigned identities. An edge case showed that assignment fields alone do not guarantee positive weight. Float32, tie, eligibility, clipping, ordering, margin, and bracket rules followed the frozen contract; zero-row and asymmetric-key rehearsal was not covered and later caused the lockbox failure.

Table S6: Instrument qualification summary.

Check	Cases/states	Mismatches	Result
Analytical oracles	6	0	pass
Property invariants	7	0	pass
Prespecified mutations	6	0 undetected	pass
Native/tracer/NumPy parity	5,551 states	0	pass
Assigned/target-score semantics	13,238 assigned GT states	0 nonpositive weights	pass
Legacy-schedule NumPy reference	4,408 trajectories	0	pass
Full-grid focal-row/native replay	3,072 scenarios	0	pass
Forward-call contract	30 images	0 excess calls	pass

Table S7: Executable analytical-oracle, metamorphic-property, and mutation-test detail. These checks qualify the measurement instrument and its governance; they are not scientific outcome evidence. Case names and outcomes are transcribed from the frozen qualification artifacts.

Case / defect name	Expected	Observed	Pass
Analytical oracle cases			
`O1_within_set`	`within_set_geometry_rank_reversal`	`within_set_geometry_rank_reversal`	Yes
`O2_eligibility`	`eligibility_boundary`	`eligibility_boundary`	Yes
`O3_top7`	`topk_membership_transition`	`topk_membership_transition`	Yes
`O4_conflict`	`conflict_reassignment`	`conflict_reassignment`	Yes
`O5_clipping`	`within_set_geometry_rank_reversal`	`within_set_geometry_rank_reversal`	Yes
`O6_disappearance`	`eligibility_boundary`	`eligibility_boundary`	Yes
Metamorphic properties			
`candidate_permutation_invariant`	`true`	`true`	Yes
`gt_order_invariant`	`true`	`true`	Yes
`direction_processing_order_invariant`	`true`	`true`	Yes
`equal_p_ranking_is_ciou_ranking`	`true`	`true`	Yes
`equal_ciou_ranking_is_p_ranking`	`true`	`true`	Yes
`conflict_off_eliminates_conflict_label`	`true`	`true`	Yes
`eligibility_lock_eliminates_eligibility_label`	`true`	`true`	Yes
Mutation cases			
`M1_remove_ciou_clip`	unclipped negative CIoU produces nonzero q where the frozen contract requires zero	`killed=true`	Yes
`M2_topk_6_or_8`	Top-k membership parity changes relative to native Top-7	`killed=true`	Yes
`M3_corrupt_target_gt_idx`	exact native/reference `target_gt_idx` parity detects the injected ownership error	`killed=true`	Yes
`M4_disable_stal_expansion`	tiny-object eligibility mask shrinks relative to the native STAL contract	`killed=true`	Yes
`M5_impute_missing_runner_as_zero`	margin-domain invariant rejects a finite value when no positive runner-up exists	`killed=true`	Yes
`M6_disable_conflict`	conflict-stage parity differs and conflict-labelled transitions are impossible	`killed=true`	Yes

4. Endpoint stress-contract results

4.1. Primary fixed and equivalent-area-side contrasts

Table S8: AI-TOD-v2 primary fragility rates and scale contrasts in percentage points. Gaps are 8–16 minus 16–32 px; the paired gap is O2O minus O2M. Brackets are 95% stratified image-cluster bootstrap intervals.

Stress contract	O2O 8–16	O2O 16–32	O2O gap	O2M 8–16	O2M 16–32	O2M gap	Paired gap
Fixed 1 px	22.14 [19.96, 24.53]	7.77 [4.91, 11.45]	14.37 [10.84, 17.78]	71.74 [69.69, 73.71]	70.32 [64.73, 75.35]	1.42 [-3.18, 6.31]	12.95 [6.52, 19.16]
Equivalent-area side $\kappa = 0.0625$	9.11 [7.86, 10.49]	8.98 [5.84, 13.03]	0.13 [-3.41, 3.15]	57.93 [55.59, 60.31]	75.37 [69.30, 80.48]	-17.45 [-22.72, -11.71]	17.58 [11.06, 23.98]

4.2. Magnitude and dataset sensitivity

Table S9: Equivalent-area-side stress sensitivity. Values are 8–16 minus 16–32 px contrasts in percentage points with 95% stratified image-cluster bootstrap intervals.

Dataset	κ	O2O	O2M	O2O−O2M
AI-TOD-v2	0.03125	1.34 [−0.72, 3.10]	−16.85 [−22.30, −11.00]	18.19 [12.48, 23.83]
AI-TOD-v2	0.06250	0.13 [−3.41, 3.15]	−17.45 [−22.72, −11.71]	17.58 [11.06, 23.98]
AI-TOD-v2	0.12500	8.91 [4.21, 13.55]	−8.36 [−11.33, −5.41]	17.27 [11.43, 23.45]
VisDrone	0.03125	5.14 [3.93, 6.36]	−29.65 [−31.73, −27.59]	34.79 [32.19, 37.42]
VisDrone	0.06250	11.04 [9.33, 12.70]	−26.21 [−27.85, −24.57]	37.25 [34.73, 39.85]
VisDrone	0.12500	19.97 [17.73, 22.25]	−16.01 [−17.17, −14.84]	35.98 [33.39, 38.56]

4.3. Direction-normalised rectangular-box sensitivity

Direction-normalised replay used $\delta_x = \kappa w$ horizontally and $\delta_y = \kappa h$ vertically. The comparison with equivalent-area-side replay used the identical 6,377-GT common-valid support and 5,000 paired stratified image-cluster bootstrap replicates.

Table S10: Equivalent-area-side versus direction-normalised scale contrasts. Observed rows use each contract's canonical formal summary. Aspect-standardised rows are a separate paired composition sensitivity that applies a common pooled aspect-ratio distribution to both size groups.

Displacement	Composition	O2O gap (pp)	O2M gap (pp)	Paired gap (pp)
Equivalent-area side	Observed	0.13 [−3.41, 3.15]	−17.45 [−22.72, −11.71]	17.58 [11.06, 23.98]
	Aspect-standardised	−0.12 [−3.01, 2.75]	−17.64 [−22.90, −11.99]	17.53 [11.07, 24.08]
Axis-normalised	Observed	2.79 [0.44, 4.96]	−17.04 [−22.16, −11.58]	19.83 [13.65, 25.83]
	Aspect-standardised	2.62 [0.45, 4.80]	−17.10 [−22.23, −11.64]	19.72 [13.67, 25.77]

The primary and canonical equivalent-area rows share point estimates and support; interval differences reflect separately executed bootstrap analyses.

Axis-minus-equivalent changes were +2.66 pp [1.10, 4.45] for the object-level O2O gap and +0.41 pp [−1.17, 1.92] for O2M, with the largest O2O change among elongated objects. The direction-level paired change was +0.48 pp [−0.31, 1.29].

4.4. Archived-stage and detector-contract sensitivity

YOLOv10-S equivalent-area-side replay yielded O2O +0.94 pp [−1.85, 3.61], O2M −23.50 pp [−28.18, −18.90], and paired +24.44 pp [18.89, 30.06].

4.5. Crossed training-seed and image sensitivity

Seeds 0–3 satisfied the frozen common contract and were included; seed 4 was excluded before outcome analysis. All included artifacts record epoch 280. The strict cross-seed intersections retained 6,318 and 6,317 GTs from 288 contributing images, with seed and stratified-image resampling.

The fixed paired contrast ranged from 12.44 to 13.19 pp, with an equal-seed mean of 12.75 pp [6.27, 18.84]. The equivalent-area paired contrast ranged from 17.20 to 19.18 pp, with an equal-seed mean of 17.94 pp [11.51, 24.23]. Equivalent-area O2M gaps were negative in every seed, and both assigned-set Top-1 turnover and positive-set turnover were higher for 16–32 px objects in every seed. Eligibility, pathway, and dense-boundary results remain conditional on seed 0.

Table S11: Aspect-stratified scale contrasts under the two displacement parameterisations (pp), with raw counts, inverse-inclusion-weighted support and 95% paired stratified image-cluster bootstrap intervals.

Aspect stratum	Raw n (8–16/16–32)	IPW support (8–16/16–32)	Eq.-area O2O gap	Axis O2O gap	Eq.-area O2M gap	Axis O2M gap
Ratio ≤ 1.5	2402/527	105863.05/23250.07	3.96 [0.91, 6.88]	5.27 [2.51, 7.92]	−18.04 [−24.18, −11.07]	−16.85 [−22.90, −10.05]
$1.5 <$ ratio ≤ 2.5	2414/363	106400.39/16013.22	0.82 [−2.86, 3.97]	1.95 [−1.05, 4.62]	−16.21 [−24.65, −8.48]	−16.78 [−24.89, −9.31]
Ratio > 2.5	570/101	25117.10/4457.17	−21.79 [−30.27, −10.00]	−6.19 [−13.01, 0.94]	−21.82 [−30.52, −11.74]	−19.49 [−30.03, −10.52]

Table S12: Paired O2O-minus-O2M scale contrasts across the complete physically archived trajectory. Epoch 0 denotes the checkpoint written after the first completed training epoch. The row labelled 300 is the archived final `last.pt` training state and is distinct from the retained post-training-selected epoch-280 artifacts named `best.pt`. No intermediate archive existed for epochs 100–280.

Epoch	$\mathrm{mAP}_{50:95}$	Fixed pp	Equivalent-area-side pp	Common-valid GTs (fixed/eq.-area)
0	0.112	13.42 [7.57, 20.12]	18.41 [11.97, 25.51]	6343/6342
20	0.191	13.06 [6.68, 19.71]	19.27 [12.92, 26.00]	6394/6393
40	0.226	15.05 [8.71, 21.35]	21.35 [14.38, 28.38]	6392/6391
60	0.243	15.09 [10.23, 20.05]	21.00 [15.65, 26.48]	6387/6386
80	0.254	14.16 [8.21, 20.12]	19.31 [13.93, 25.03]	6394/6393
300	0.280	11.84 [5.70, 18.10]	17.34 [11.22, 23.38]	6374/6373

5. Eligibility, first-divergence, and pre-resolution controls

5.1. Eligibility-lock and eligibility-stable estimands

On the lock support, holding base eligibility fixed reduced the fixed O2O gap from 17.10 to 3.52 pp and the equivalent-area gap from 2.95 to −0.20 pp; the primary common-valid estimates use separate supports (6,378 and 6,377 GTs).

The post-replay eligibility-stable subset retained 6,217/7,434 fixed and 6,999/7,432 equivalent-area objects. Object-level contrasts were 3.36 pp [1.11, 5.48] and 1.05 pp [−1.19, 2.96], respectively; direction-level contrasts were 1.02 pp [0.19, 1.74] and 0.02 pp [−0.86, 0.76]. These are selected estimands.

The four-valid restriction retained 96.87–98.70% of common-valid objects across scale groups and contracts. It changed the fixed paired gap from 12.95 to 12.59 pp and the equivalent-area gap from 17.58 to 17.24 pp; both four-minus-all intervals included zero.

5.2. First-divergence pathway anatomy

The historical taxonomy labels "Active disappearance" and "compound/other" are fallback first labels used only when no earlier eligibility, Top-k, conflict, or within-set category applies; zero first-label rows do not exclude later occurrence on the same path.

Within-set rank changed somewhere on 75.12% of fixed identity-exchange paths but was first on 30.13%; 44.63% also contained eligibility change. The equivalent-area values were 77.44%, 45.37%, and 31.59%. First-label shares depend on stage order.

Frozen outputs make the stress-induced classification term zero. The decomposition therefore separates clipped-CIoU movement among comparable candidates from discrete eligibility, Top-k, and conflict transitions. Algebraic residual below 2×10^{-7} served as an implementation check.

5.3. Pre-resolution O2O control

Pre-conflict versus final assigned scale gaps were 17.18 versus 17.10 pp under fixed replay and 3.34 versus 2.95 pp under equivalent-area replay. Their differences were −0.077 pp [−0.367, 0.313] and −0.395 pp [−0.986, 0.034]. Object-level disagreement was 0.578% under both contracts, and every directional disagreement coincided with a conflict-stage difference.

6. Continuous cardinal-direction boundary persistence

The formal analysis evaluated 29,724 trajectories from 7,431 focal GTs at every legal 1/1024 point through $r_{\max} = 0.25$ on four cardinal rays, retaining returns, right censoring, and competing changes. Raw event, cause, and return reports remain separate; the grid is not a two-dimensional minimum and cannot resolve sub-step intervals.

The 1/64 plus 1/512 schedule was rejected after a narrow-state mutation and a 30-image comparison exposed nine material distance or return-state mismatches; legacy outputs are aliasing diagnostics only.

Table S13: Fixed-one-pixel scale contrasts across four protocol-eligible training seeds on the strict cross-seed common-valid support. The equal-seed mean interval uses 20,000 crossed seed and stratified image-cluster bootstrap replicates. With four top-level seeds, the interval remains a sensitivity analysis rather than precise detector-training-level uncertainty.

Training seed	O2O gap (pp)	O2M gap (pp)	Paired gap (pp)
0	14.22	1.75	12.47
1	15.01	2.57	12.44
2	14.19	1.00	13.19
3	15.13	2.25	12.88
Equal-seed mean	14.64 [11.78, 17.60]	1.89 [−2.96, 7.34]	12.75 [6.27, 18.84]

Table S14: Equivalent-area-side endpoint contrasts across the same four seeds on the strict cross-seed common-valid support. The equal-seed mean interval uses 20,000 crossed seed and stratified image-cluster bootstrap replicates.

Contrast (pp)	Seed 0	Seed 1	Seed 2	Seed 3	Equal-seed mean [95% CI]
O2O, 8–16 minus 16–32	0.03	1.39	0.78	1.42	0.90 [−2.11, 3.58]
O2M, 8–16 minus 16–32	−17.17	−16.13	−17.10	−17.77	−17.04 [−22.35, −11.27]
O2O minus O2M	17.20	17.51	17.87	19.18	17.94 [11.51, 24.23]

At $\tau = 0.125$, O2O boundary incidence was 9.05% and 6.39% versus 32.75% and 48.25% for O2M across the two size groups; RMCBD contrasts were 0.01745 and 0.03319. Return-to-base rates were 0.049% and 0.721%, respectively.

Across the six evaluated truncation horizons, the O2O-minus-O2M contrast remained positive. Each branch-specific RMCBD is nondecreasing in τ by construction; the primary $\tau = 0.125$ estimate is one prespecified horizon.

For fixed-pair rank-crossing distance ρ_R, 1,593 trajectories were observed crossings, 7,439 were competing-censored, 16,548 were right-censored, and 4,144 were undefined. Undefined reasons were 676 base-assigned-identity missing, 56 base order not established over runner, and 3,412 base runner missing.

7. O2M Rank–Set and same-stride results

The legacy eligible Top-1 field is inactive. Reported O2M Top-1 results use the maximal-q member of the post-conflict assigned-positive set, a rank probe within multi-positive supervision.

Equivalent-area positive counts were 3.07 and 6.54 across the two size groups, with a base-margin difference of −0.104. Same-stride local, nonlocal, and cross-stride transitions accounted for 55.0%, 21.6%, and 23.3% of the reverse directional gap; no fixed-positive-count matched estimate was produced.

7.1. Count-conditioned and continuous-size controls

The adequate-overlap target covered 60.29% and 64.30% of the two weighted scale groups. The table reports the object-level sequences 17.45, 4.53, and 2.16 pp; direction-level sequences 11.59, 3.59, and 0.88 pp; and positive-set-turnover sequences 33.16, 36.82, and 41.13 pp. Without complete ordered q scores, the archived tables cannot reconstruct fixed-K replay.

Alternative cutpoints and continuous log-area models used 6,376 object and 25,371 directional rows from 288 images with the recorded covariates; Table S24 reports the descriptive associations. Legacy-schedule boundary associations are excluded.

The strict post-replay same-stride subset retained 3,835 paired GTs. Its O2O contrast was +10.23 pp [7.30, 13.32], O2M was −0.12 pp [−6.83, 5.78], and the paired difference was +10.35 pp [3.54, 18.17]. This sensitivity rules out an explanation based solely on cross-stride switching but remains vulnerable to selection induced by post-replay conditioning.

8. Analysis support map and computational context

Table S25 reports analysis-specific denominators; "selected/contributing" distinguishes the frozen sample from images supplying at least one row.

All rows map to `evidence_freeze/final_evidence_v4`. The four-seed sensitivity records are registered as EV-MULTISEED-FIXED-COMPLETION-V1 and EV-MULTISEED-EQUIVALENT-RANKSET-V1; the immutable `evidence_freeze/final_evidence_v3` predecessor is preserved.

Table S15: Equivalent-area-side Rank–Set and native positive-count contrasts across the same four seeds. Each seed contributes 25,165 legal directions from the 6,317-GT strict endpoint intersection. Intervals use the crossed seed and stratified image-cluster bootstrap.

16–32 minus 8–16 contrast	Seed 0	Seed 1	Seed 2	Seed 3	Equal-seed mean [95% CI]
Native positive count	3.45	3.45	3.44	3.45	3.45 [3.17, 3.75]
Jaccard (pp)	−8.10	−8.12	−8.01	−8.05	−8.07 [−8.78, −7.38]
Base retention (pp)	−4.94	−4.94	−4.82	−4.84	−4.88 [−5.26, −4.49]
Assigned-set Top-1 flip (pp)	11.49	12.26	11.58	12.35	11.92 [8.62, 15.15]
Positive-set turnover (pp)	33.23	33.17	33.21	32.83	33.11 [31.32, 34.76]
Rank-only among flips (pp)	−31.68	−30.18	−27.88	−29.79	−29.88 [−33.43, −26.38]

Table S16: Sensitivity to requiring all four cardinal shifts to be legal among primary common-valid O2O/O2M objects. “All legal” is the original any-observed-legal-direction object estimand; “four-valid” excludes objects with fewer than four legal directions. Difference intervals use 5,000 paired stratified image-cluster bootstrap replicates.

Contract	Estimand	All legal (pp)	Four-valid (pp)	Four-minus-all (pp)
Fixed 1 px	O2O gap	14.37	14.39	0.02 [−0.19, 0.28]
	O2M gap	1.42	1.80	0.38 [−0.10, 0.77]
	Paired gap	12.95	12.59	−0.37 [−0.77, 0.14]
Equivalent-area side	O2O gap	0.13	0.18	0.05 [−0.27, 0.48]
	O2M gap	−17.45	−17.06	0.39 [−0.03, 0.74]
	Paired gap	17.58	17.24	−0.34 [−0.80, 0.19]

The local workstation used Windows 11, an Intel Core i7-14650HX CPU, 15.6 GiB RAM, and an NVIDIA RTX 5060 Laptop GPU with 8,151 MiB. Software comprised Python 3.13.5, PyTorch 2.11.0+cu128, CUDA 12.8, and modified Ultralytics 8.4.51. These values provide execution context.

Timing sources have SHA-256 prefixes `38299ec8fdc5`, `5e2f0e689bc3`, and `f6427ea0715b`. The three-worker duration is reconstructed from frozen start and completion records. It covers the remaining 203 images after 97 images had already completed, so it is not expected to equal one third of the 300-image cumulative image-compute total. Final event and curve hashes passed terminal validation.

9. Conditional margin alignment, shape, and held-out diagnostic value

Among 1,593 observed fixed-pair crossings, margin correlated with ρ_R at $\rho = 0.628$ [0.584, 0.669], versus 0.150 [0.111, 0.191] for observed eligibility-boundary distance; the difference was 0.477 [0.408, 0.543]. These conditional correlations are not censoring adjusted.

10. Locality, audit budget, and secondary negative boundaries

10.1. Non-focal spillover

Among 3,549,250 focal–non-focal pair-events, 52 identity changes were observed, for a weighted rate of 0.001465% [0.000824%, 0.002383%]. All occurred in 5,679 pairs sharing a pre-conflict claim, where the rate was 0.9159% [0.5787%, 1.3248%]; none occurred among 3,543,571 nonsharing pairs.

10.2. Reduced audit budget

At 100 images, all 1,000 repeats recovered the full-sample sign and branch ordering; median and 95th-percentile absolute errors were 3.43 and 8.41 pp.

10.3. Secondary intervention records

Historical intervention experiments are excluded because they lack one canonical checkpoint and evaluation contract. Future inclusion would require regeneration with dataset, split, checkpoint SHA-256, seed, evaluator, and experiment ID recorded per row.

Table S17: Mutually exclusive first-divergence shares among directional O2O identity exchanges. Brackets are 95% stratified image-cluster bootstrap intervals from 5,000 replicates. Geometry clipping is an overlapping descriptive diagnostic and is not included in the 100% sum.

Pathway	Fixed 1 px	Equivalent side	Equivalent minus fixed
Eligibility boundary	67.09 [63.02, 71.07]	51.01 [46.06, 55.94]	−16.09 [−18.92, −13.34]
Top-k transition	0.27 [0.05, 0.54]	0.48 [0.17, 0.85]	0.21 [−0.04, 0.48]
Conflict reassignment	2.51 [1.07, 4.12]	3.14 [1.61, 4.93]	0.64 [−0.71, 1.96]
Within-set geometry	30.13 [26.14, 34.36]	45.37 [40.62, 50.26]	15.24 [12.69, 17.83]
Active disappearance	0	0	0
Compound/other	0	0	0
Directional flip rows	2,194	1,241	n/a
Images with at least one exchange	201	159	n/a
Geometry-clipping event	44.72 [41.18, 48.73]	31.83 [27.92, 36.13]	n/a

Table S18: First-label shares versus multi-label stage occurrence among directional O2O identity exchanges (%). "Any" columns do not enforce exclusivity. Brackets are 95% stratified image-cluster bootstrap intervals from 5,000 replicates.

Contract	First eligibility	First within-set	Any eligibility	Any within-set	Eligibility ∩ within-set
Fixed 1 px	67.09	30.13	67.09 [63.01, 71.14]	75.12 [70.98, 79.35]	44.63 [41.07, 48.81]
Equivalent-area side	51.01	45.37	51.01 [45.98, 55.90]	77.44 [73.15, 81.84]	31.59 [27.78, 35.68]

11. Qualitative provenance

The stable, fragile, and undefined examples were selected from the frozen qualitative manifest before final layout. They illustrate the native state definitions; undefined examples remain excluded from continuous margin analysis without imputation.

Table S19: Full-domain 1/1024 restricted mean cardinal boundary distance (RMCBD) through $\tau = 0.125$.

Quantity	Point	95% CI
O2O RMCBD	0.11985	[0.11932, 0.12032]
O2M assigned-positive RMCBD	0.10025	[0.09864, 0.10180]
Tiny O2O–O2M RMCBD	0.01745	[0.01586, 0.01898]
Small O2O–O2M RMCBD	0.03319	[0.02937, 0.03652]

Table S20: Sensitivity of the O2O-minus-assigned-positive-O2M RMCBD contrast to the truncation point τ. Brackets are 95% stratified image-cluster bootstrap intervals from 5,000 replicates.

τ	All	8–16 px	16–32 px
0.0500	0.0038 [0.0034, 0.0043]	0.0033 [0.0030, 0.0037]	0.0072 [0.0061, 0.0083]
0.0625	0.0058 [0.0052, 0.0064]	0.0050 [0.0045, 0.0056]	0.0106 [0.0090, 0.0121]
0.1000	0.0134 [0.0122, 0.0147]	0.0119 [0.0108, 0.0130]	0.0232 [0.0203, 0.0258]
0.1250	0.0196 [0.0179, 0.0213]	0.0174 [0.0159, 0.0190]	0.0332 [0.0294, 0.0365]
0.2000	0.0413 [0.0380, 0.0445]	0.0372 [0.0340, 0.0401]	0.0673 [0.0612, 0.0727]
0.2500	0.0567 [0.0522, 0.0609]	0.0513 [0.0470, 0.0552]	0.0911 [0.0836, 0.0980]

Table S21: Tiny-minus-small O2M Rank–Set contrasts in percentage points with 95% stratified image-cluster bootstrap intervals.

Contract	ΔJaccard	Δretention	ΔTop-1 flip	Δrank-only share
Fixed 1 px	−4.19 [−4.89, −3.50]	−2.26 [−2.78, −1.78]	−2.38 [−5.14, 0.42]	−10.10 [−14.50, −6.20]
Equivalent-area side	5.91 [5.18, 6.64]	4.93 [4.54, 5.31]	−12.80 [−16.02, −9.33]	22.50 [18.39, 26.33]

Table S22: O2M empty-set structural transitions with 95% stratified image-cluster bootstrap intervals. Empty/empty has undefined Jaccard and retention; empty/nonempty and nonempty/empty remain explicit state changes.

Contract	Structural state	8–16 px (%)	16–32 px (%)	Tiny-minus-small (pp)
Fixed 1 px	Empty → empty	1.80 [1.21, 2.59]	0	1.80 [1.21, 2.59]
	Empty → nonempty	1.02 [0.79, 1.28]	0	1.02 [0.79, 1.28]
	Nonempty → empty	1.07 [0.82, 1.38]	0.03 [0, 0.08]	1.05 [0.80, 1.35]
Equivalent-area side	Empty → empty	1.83 [1.24, 2.62]	0	1.83 [1.24, 2.62]
	Empty → nonempty	0.99 [0.76, 1.24]	0	0.99 [0.76, 1.24]
	Nonempty → empty	0.04 [0.01, 0.07]	0.03 [0, 0.08]	0.01 [−0.05, 0.06]

Table S23: O2M positive-count controls on common-valid equivalent-area-side support. Exact-$K = 3$–7 is the adequate-overlap set with at least 20 raw GTs from each size group at every retained count. Full-support crude, overlap-support crude, and overlap-support standardised estimates separate support restriction from count-composition standardisation. Intervals use 5,000 stratified image-cluster bootstrap replicates.

Outcome (16–32 minus 8–16)	Full crude	$K = 3$–7 overlap crude	$K = 3$–7 standardised
Object-level assigned Top-1 fragility	17.45 [11.85, 22.73]	4.53 [−1.46, 10.52]	2.16 [−4.74, 9.71]
Direction-level assigned Top-1 flip	11.59 [8.42, 14.66]	3.59 [0.80, 6.63]	0.88 [−2.18, 4.47]
Direction-level positive-set turnover	33.16 [31.35, 34.79]	36.82 [34.73, 38.87]	41.13 [38.23, 44.26]

Table S24: Cutpoint and continuous-size controls on common-valid equivalent-area-side support. Panel A reports upper-minus-lower fragility gaps at cutpoint c among objects with equivalent-area side in $[8, 32)$ px. Panel B reports adjusted probability changes per area doubling from sampling-weighted linear probability models. Intervals use 2,000 stratified image-cluster bootstrap replicates.

Analysis	Contrast or outcome	Estimate (pp)	95% CI (pp)
Panel A: upper-minus-lower fragility at alternative equivalent-area-side cutpoints			
$c = 14$ px	O2O fragility	−1.15	[−3.21, 1.02]
	O2M fragility	16.26	[11.41, 20.91]
$c = 16$ px	O2O fragility	−0.13	[−3.23, 3.31]
	O2M fragility	17.45	[11.85, 22.73]
$c = 18$ px	O2O fragility	−0.15	[−3.77, 4.00]
	O2M fragility	19.74	[13.90, 25.50]
Panel B: adjusted probability change per area doubling			
Continuous model	Object-level Top-1 fragility	7.11	[3.28, 10.84]
	Rank–Set turnover	15.75	[13.92, 17.59]

Table S25: Support map for the principal and sensitivity analyses. Bootstrap counts refer to stratified image-cluster resampling unless another top-level unit is stated.

Analysis	Image support	Analysis units	Support restriction	Bootstrap
Primary fixed endpoint	300/288	6,378 GT	Contract-specific O2O/O2M common-valid	5,000 images
Primary equivalent-area endpoint	300/288	6,377 GT	Contract-specific O2O/O2M common-valid	5,000 images
Eligibility lock	300/300	7,434/7,432 GT; 29,616/29,612 directions	All legal fixed/equivalent-area focal GT; native and locked compared within arm	5,000 images
Eligibility-stable objects	300/282 fixed; 300/293 equivalent	6,217/6,999 GT	Candidate-pair eligibility stable through all legal replay states	5,000 images
First-divergence anatomy	300/201 fixed; 300/159 equivalent	2,194/1,241 exchange directions	Directional O2O identity exchanges only	5,000 images
Rank–Set	300/300	7,434/7,432 GT; 29,616/29,612 directions	Native post-conflict O2M assigned-positive states	5,000 images
Strict same-stride	300/243	3,835 paired GT	Both margins valid; every legal replay retains base stride in both branches	5,000 images
Dense-grid KM/AJ/RMCBD	300/300	7,431 GT; 29,724 trajectories	Thirteen focal GTs excluded from the dense-grid analysis domain by the no-clipping contract	5,000 images
Exact endpoint/return	300/300	28,801 O2M directions	Legal replay state exists at exact $\tau = 0.125$ endpoint; no interpolation	5,000 images
Margin–ρ_R	300/173	1,593 observed crossings	Observed fixed-pair crossings only; not censoring adjusted	5,000 images
YOLOv10-S replication	300/288	6,348 GT	Compatible-detector common-valid support	5,000 images
Four-seed fixed sensitivity	300/288	6,318 GT	Cross-seed O2O/O2M common-valid intersection; seeds 0–3	20,000 seed + image
Four-seed equivalent/Rank–Set	300/288	6,317 GT; 25,165 directions/seed	Cross-seed endpoint intersection; seeds 0–3	20,000 seed + image

Table S26: Recorded audit wall times. Endpoint manifests did not retain run-specific CPU, RAM, or VRAM peaks, so no utilisation claim is made.

Audit	Recorded time	Interpretation
Fixed endpoint, 300 images	595.9 s replay; 619.9 s total	7,434 GT and 29,616 direction rows on device 0
Fixed pathway anatomy, 300 images	342.4 s	7,434 GT through the instrumented first-divergence path
Full 1/1024 boundary grid	52,944.6 s cumulative image compute	Sum across the completed image shards; not user waiting time
Three-worker boundary phase	approximately 3 h 13 min 49 s wall time	Earliest worker start to latest completion; workers executed concurrently

Table S27: Held-out effect of replacing the linear margin term with a restricted cubic spline. Values are spline minus linear model.

Outcome	Metric	Difference	95% CI
Fragility	ROC-AUC	−0.00756	[−0.01411, −0.00138]
Fragility	PR-AUC	−0.00415	[−0.00813, −0.00109]
Fragility	Log loss	−0.00090	[−0.00260, 0.00075]
Coverage-based miss	ROC-AUC	−0.00057	[−0.00122, 0.00002]
Coverage-based miss	PR-AUC	−0.00541	[−0.01298, −0.00027]
Coverage-based miss	Log loss	−0.00060	[−0.00794, 0.00526]

Table S28: Held-out diagnostic change after adding margin by dataset. The outcome is the coverage-based miss: no frozen same-class prediction among the validation outputs at confidence 0.001 and `max_det`=300 overlaps the focal GT at IoU ≥ 0.5. Five image-grouped folds compare M3 (M2 plus margin) with M2 (log area, log O2M positive count, log assigned score, and assigned clipped CIoU). Baseline values are fold-weighted M2 metrics; intervals use 5,000 image-cluster bootstrap replicates within out-of-fold fold and sampling stratum. Differences are M3 minus M2, so negative log-loss and Brier differences indicate improvement.

Dataset	Metric	M2 baseline	M3−M2	95% CI
AI-TOD-v2	ROC-AUC	0.95354	−0.00100	[−0.00484, 0.00036]
AI-TOD-v2	PR-AUC	0.79549	0.00366	[−0.00405, 0.01197]
AI-TOD-v2	Log loss	0.14653	0.00041	[−0.00362, 0.00534]
AI-TOD-v2	Brier	0.039508	0.000012	[−0.000375, 0.000496]
VisDrone	ROC-AUC	0.96747	−0.00041	[−0.00148, 0.00000]
VisDrone	PR-AUC	0.86613	−0.00044	[−0.00161, 0.00001]
VisDrone	Log loss	0.18984	0.00256	[−0.00005, 0.00807]
VisDrone	Brier	0.056757	0.000066	[−0.000011, 0.000221]

Table S29: Outcome-blind subsampling error for the equivalent-area-side paired contrast; 1,000 repeated stratified subsamples per budget.

Budget	Median error	P95 error	Sign recovery	Order recovery
25	6.39	18.38	97.5%	97.5%
50	4.52	13.63	99.6%	99.6%
100	3.43	8.41	100.0%	100.0%
150	2.45	6.13	100.0%	100.0%
200	1.74	4.60	100.0%	100.0%

Figure S1: Audited primary-checkpoint assignment-state examples. Each row shows a wider context, the base state, and the one-pixel shift state for stable, fragile, and post-shift undefined cases. Green solid boxes mark the focal GT, red dashed boxes the shifted GT, blue solid boxes the assigned O2O candidate, and purple dashed boxes the runner candidate. Image pixels are unchanged and only the focal-GT centre moves by one input pixel. Detail insets reproduce source pixels without interpolation or enhancement. These overlays are audited assignment states, not final detections.

12. Historical prospective-audit record and analyzer QA

The historical prospective audit used a disjoint outcome-blind 300-image sample with the implementation, numerical contract, sampling, taxonomy, bootstrap, checkpoint hashes, and Tier-A/Tier-B rules frozen before access. Four sealed extraction roles completed; the single permitted analyzer invocation wrote the access receipt and stopped before any H1–H5 estimate or decision. It was not rerun.

Before access, 5,000 stratified image-cluster bootstrap replicates over the 300 discovery images planned expected precision for five prespecified contrasts. Table S30 reports discovery estimates, expected 95% half-widths, and direction recovery as planning diagnostics rather than post-hoc power or guarantees.

Table S30: Prospective precision planning from the 300-image discovery sample. Expected half-width is half the percentile 95% bootstrap interval. Direction recovery is the proportion of discovery bootstrap replicates matching the prespecified sign. No lockbox outcome was used.

Hypothesis	Discovery estimand	Point (pp)	95% half-width (pp)	Direction recovery
H1	Eq.-area paired gap	17.58	6.56	100%
H2	Eq.-area O2M gap	−17.45	5.53	100%
H3	Fixed minus eq.-area O2O gap	14.24	1.73	100%
H4a	Fixed eligibility minus eq.-area	16.09	2.78	100%
H4b	Eq.-area within-set minus fixed	15.24	2.58	100%

No hypothesis received a statistical decision and no normative verdict exists. The post-access diagnostic retains formal status `null`, is exploratory, and remains outside the scientific claim ledger.

Table S31: Post-access diagnostic contrasts retained for audit-trail completeness. These exploratory estimates are inadmissible as confirmatory evidence, excluded from the scientific claim ledger, and not preregistered verdicts.

Analogue	Point	Exploratory 95% CI	Formal status
H1	16.69	[12.47, 20.74]	invalidated; not retested
H2	-16.99	[-20.62, -13.34]	invalidated; not retested
H3	14.00	[12.51, 15.60]	invalidated; not retested
H4	21.15	[18.54, 23.44]	invalidated; not retested
H5	19.31	[16.53, 21.94]	invalidated; not retested

The diagnostic retained all 300 clusters and the 7,243-key common-valid intersection, with three fixed-only keys disclosed; it remains exploratory and does not restore prospective status. The failure motivates pre-outcome testing of zero-row clusters, asymmetric keys, empty strata, all-zero outcomes, duplicates, merge order, and support assertions.

12.1. Post-access analyzer repair and regression validation

The analyzer was repaired without changing Lockbox v1. It passed 33 self-tests and a 300-cluster full-CLI rehearsal covering zero-row clusters, an empty stratum, all-zero outcomes, unequal rosters, asymmetric GT keys, row-order permutation, and duplicate rejection; the synthetic audit contained 597 valid keys per arm, a 594-key intersection, and three asymmetric keys on each side, with duplicates failing closed.

A separate read-only parity check reproduced all five exploratory estimates, intervals, empirical p-values, and 5,000 bootstrap trajectories, with maximum absolute difference 9.72×10^{-17}. Support counts were 7,246 fixed-valid and 7,243 equivalent-area-valid keys, with three fixed-only keys. The access-receipt SHA-256 prefix remained `8a5eaee6703c`, and no `LOCKBOX_VERDICT.json` was created. These are software-QA checks only; all post-access estimates remain exploratory with formal status `null`.

13. Reproducibility inventory

The article-matched GitHub release v3.2.0 provides the released code, derived analysis materials, complete (1/1024) dense-grid records, and checkpoints. The corresponding article source package is preserved in the Zenodo record 10.5281/zenodo.22478277. Release assets are bound to the SHA-256 manifests under AGPL-3.0-only. Raw AI-TOD-v2 and VisDrone images and annotations are not redistributed and must be obtained from their providers; raw prediction dumps are not redistributed.

Table S32: Minimum file-level audit trail for the final manuscript.

Component	Frozen record
Evidence authority	`evidence_freeze/final_evidence_v4`; the v3 bundle is retained as the immutable predecessor
Numerical manuscript values	`manuscript/generated_final_v2` and the canonical dense-grid post-processing root `runs/20260903_boundary_full_grid_1024_postprocess_v2`
Boundary science freeze	`results/measurement_validation_20260903/dense_grid_science_freeze_v1` with the five prespecified structural decisions and machine-readable assessment
Training/checkpoint contract	Public v3.2.0 asset registry plus v4 checkpoint-provenance audit, checkpoint-embedded training arguments, dataset template, and full-validation recheck records
Prospective record	Immutable access receipt, failed analyzer log, and no-verdict record
Post-access repair QA	`results/measurement_validation_20260904/lockbox_analyzer_synthetic_cli_e2e_v2` and `results/measurement_validation_20260904/lockbox_repaired_analyzer_post_access_parity_v1`; software QA only
Public materials	GitHub v3.2.0: complete dense curve, four checkpoints, and SHA-256 records; corresponding article source package preserved in Zenodo 10.5281/zenodo.22478277.